



Application Note

Vantage Example Scripts – NDE_NDT

Summary

This documentation provides a matrix summary listing all available example scripts f in the directory of Examples_NDE_NDT. In addition, a brief description of each example type, and associated with the first two columns of the matrix (“File Name” and “Description of Example”), as well as Specialty_Applications, are provided afterwards. These notes are extracted from the documentation in the header in the example script source code.

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Imaging Mode programming example scripts

(based on Imasonic 5 MHz probe with UTA-160-SH/DH)

FlashAngles, Plane Waves with multiple angles

Sequence programming file for Imasonic 5MHz 64 el Phase array, using plane wave compounding. For each acquisition, all channels are active for transmit and receive. Reconstruction and processing can be set to synchronize with respect to acquisition P.sync.

FMC, Full Matrix Capture

Sequence programming file for Imasonic 5MHz 64 el Phase array, using total focusing reconstruction. For each acquisition, one channel is active for transmit and 64 channels are active for receive. Reconstruction and processing are synchronized with respect to acquisition.

LineMode, Conventional Scanlines with Focused Beams

Sequence programming file for Imasonic 5MHz 64 el Phase array, using 64 ray lines (focused transmits) and receive acquisitions. For each acquisition, all channels are active for transmit and receive. Reconstruction and processing can be set to synchronize with respect to acquisition P.sync.

WideBeam, Weakly Focused Overlapping Beams

Sequence programming file for Imasonic 5MHz 64 el Phase array, using overlapping wide beams. For each acquisition, all channels are active for transmit and receive. Reconstruction and processing can be set to synchronize with respect to acquisition P.sync.

FMCDataviewer, Full Matrix Capture with GUI based parameters control

Sequence programming example of full matrix capture of RF data with real-time data display, and system parameters control based on GUI. One element transmits at a time and display the RF data received on the same element. FFT of the RF data is also displayed for spectrum analysis. A number of transmit, receive and processing parameters can be updated in runtime on GUI to see the immediate effect on RF data. This script also provides an example of custom GUI development.

LambWaveDispersion

SetUpLambWaveDispersion

Sequence programming file for mode inspection of guided waves in plates. Single channel on Lemo connector is used to excite lamb waves. Other 64 channels from a hypertronics connector are allocated to detect the wavefields on an Imasonic 64 el phase array. The theorital dispersion curves are for 4mm thick alunimum plates. In experiment, the phase array should be placed such that the el 1 is closest to the lamb wave source and el 64 is furthest away from the lamb wave source. Pleaes refer to AppNote_LambWave.pdf for more details.

Verasonics Example Scripts_NDE_NDT

File Name (root)	Description of Example	UTA-160-DH / SH	UTA-160-SI	UTA-LEMO-Only
Setup_Probe_root.m		Imasonic - 5 MHz	Olympus - 5 MHz	
FlashAngles	Multiple angle plane waves/ Coherent compounding	★	★	
FMC	Full Matrix Capture (FMC)/ Synthetic Aperture	★	★	★
LineMode	Conventional Scanlines with Focused Beams	★	★	
WideBeam	Weakly Focused Overlapping Beams	★	★	
FMCDataViewer	Full Matrix Capture with RF Data Viewer	★	★	★
LambWaveDispersion				
LambWaveDispersion	Guided waves in plates	★		

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